



BGO scintillation crystals

Description: Bismuth Germanate (BGO) Scintillation crystal

Scintillation crystal	:	BGO ($\text{Bi}_4\text{Ge}_3\text{O}_{12}$)
Physical form	:	Single crystal
Melting point	:	1323 K
Chemical nature	:	BGO

Physical properties

Density	:	7.13 g/cm ³
Max emission wavelength	:	480 nm
Decay time	:	300 ns
Refractive index at emission max.	:	2.15
Photoelectron yield with bi-alkali	:	15-20% relative to NaI(Tl)
Optical attenuation coefficient	:	> 1 m
Crystal class	:	No cleavage
Useful temperature range	:	-40°C to +100°C
Storage conditions	:	no restrictions
Thermal expansion	:	7.10 ⁻⁶ K ⁻¹
Maximum thermal gradient	:	Depending on the size of a crystal For small cm size crystals: 10 degrees/min

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